

A Falsifiable Certificate Architecture for a Physical-Roof Determinant in the Equilateral Three-Disk Flow

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Abstract

This article specifies a falsifiable certificate architecture for a classical transfer determinant driven by the physical Euclidean-flight roof of the no-eclipse equilateral three-disk flow at center separation $d = 6a$. It synthesizes a frozen 26-source corpus and reports no constructed roof, operator theorem, determinant computation, numerical enclosure, fidelity result, nontransfer certificate, or novelty assessment. The evidence supports a strict type firewall among physical dynamics, symbolic suspensions, semiclassical periodic-orbit objects, exact-quantum determinants, and classical transfer determinants. It also supplies conditional components for collision coding, transfer operators, determinant identities, numerical approximation, and Livšic-type obstruction logic, but not the exact object map required for the frozen geometry. We organize the resulting obligations into six open gates: physical-roof construction, operator eligibility, a common coefficient map, a five-channel common-norm error theorem, typed physical-fidelity controls, and theorem-licensed nontransfer. The error contract distinguishes four numerical channels—orbit tail, rank or projection, quadrature or evaluation, and roundoff—from separately propagated geometry and roof-input uncertainty. No channel may be combined without explicit stability, propagation, dependency, and determinant-conditioning arguments. Internal Euler, trace, and Fredholm agreement remains roof-agnostic calibration. All citations lack passage locators, so claim-to-passage support is inconclusive. The fixed state remains AO_FAIL / A2_NOT_ELIGIBLE / NO_ROUTE_PROMOTION.

Keywords: three-disk scattering; physical roof; transfer operator; Fredholm determinant; error propagation; certificate architecture

Traditional Chinese Abstract / 繁體中文摘要

本文為無遮蔽、等邊三圓盤流在中心距離 $d = 6a$ 時的物理歐氏飛行屋頂，提出一套可否證的古典轉移行列式證書架構。研究只綜整凍結的二十六項來源，不建構點態屋頂、算子定理、行列式數值、誤差包絡、物理忠實度結果或非轉移證書，也不進行新穎性判定。證據要求嚴格區分物理動力學、符號懸掛、半古典週期軌道、精確量子行列式與古典轉移行列式；相同幾何或編碼不能使不同型別彼此替代。本文將未完成義務整理為六道開放關卡：物理屋頂建構、算子適格性、共同係數映射、五通道共同範數誤差定理、型別化物理控制，以及由定理授權的方向性非轉移推論。誤差契約把軌道尾端、秩或投影、求積或評估及捨入四項數值誤差，與幾何及屋頂輸入不確定性分開傳遞；若缺少穩定性、傳播、相依性與行列式條件控制，便不得宣稱誤差可加、完備或已形成包絡。Euler、週期跡與 Fredholm 行列式的內部一致僅是與屋頂型別無關的校準，不證明物理忠實度。所有引文均缺段落定位，故支持狀態仍不確定；既定路線狀態維持失敗、無資格且不晉級。

關鍵詞： 三圓盤散射；物理屋頂；轉移算子；Fredholm 行列式；誤差傳播；證書架構

1 Introduction, Research Question, and Contribution Boundary

The frozen physical object is the planar, no-eclipse, equilateral three-disk flow with disk radius a and center separation $d=6a$. The clock is Euclidean free-flight time between collisions. Prospective owners are realized primitive cyclic collision words; traversal powers are repetitions. The pointwise roof must be calculated from collision geometry and the registered section, not fitted from inherited periodic totals. A unit roof is a typed symbolic control, not the physical clock. These are project definitions and do not emerge as new conclusions from the present synthesis.

The research question asks what must be certified before a classical nonconstant-roof transfer determinant can be interpreted as belonging to that exact physical flow. Shared geometry and periodic-orbit vocabulary create a particular risk: physical, semiclassical, exact-quantum, and classical transfer objects can be discussed with similar symbols while carrying different maps, weights, domains, and spectral meanings. The study therefore begins with an object-type firewall rather than with determinant values.

The proposed answer is a six-gate program. A future implementation would have to construct the physical roof, prove operator eligibility, prove a common coefficient map, establish a five-channel common-norm error theorem, validate physical fidelity against typed controls, and license a directional nontransfer inference. Each gate has a stop state, and none is closed in this article. The program is intended to make a later numerical claim falsifiable rather than to imply that a determinant already exists.

The error gate receives special emphasis because a list of numerical tolerances is not an enclosure. Four numerical components are prospectively distinguished: orbit truncation, finite-rank or projection approximation, quadrature or determinant evaluation, and roundoff. A fifth and separately propagated channel is the uncertainty in geometry and the pointwise roof supplied to the operator. These heterogeneous channels may be combined only after a common norm, stability map, propagation rule, and determinant-conditioning bound are proved. The present architecture does not call their sum a total error, claim exhaustiveness, or assume additivity.

The contribution is a nonarithmetic certificate architecture for classical dynamics and validated computation. It makes no determinant, fidelity, cohomology, or Route result; internal physical-roof coherence cannot repair absent Route-A arithmetic specificity.

1.1 Typed claim graph and falsifiability

The central discipline is to bind every assertion to a typed object. The physical flow is determined by disk geometry, the collision section, and Euclidean flight time. A symbolic suspension consists of a shift or return map together with a roof; changing that roof changes the timed flow even when the admissible words are unchanged. A semiclassical construction may use the same periodic rays while assigning phase, stability, or wave-number weights. An exact-quantum determinant refers to a different operator and spectral question. A classical transfer determinant again requires its own operator, potential, space, normalization, and analytic domain. Similar notation and a shared orbit vocabulary do not erase these differences.

This graph separates three levels of agreement. Formal agreement means that Euler, repetition, trace, and determinant expressions have been expanded using the same declared weights and conventions. Analytic agreement additionally requires a lawful operator and a theorem relating its traces or spectrum to the determinant on a stated domain. Finite numerical agreement compares declared approximations under an explicit tolerance. Each level can reveal an implementation inconsistency, but none by itself identifies the roof as the Euclidean-flight roof of the frozen geometry. Physical identity requires an independent geometry-to-roof construction and validation against geometry-derived periodic data.

Falsifiability enters through stop states rather than through an expectation of success. If the geometry cannot produce a pointwise roof without fitting inherited totals, Gate 1 stops. If a proposed function space

does not satisfy the required operator hypotheses, Gate 2 stops. If orbit and projected-operator coefficients cannot be connected under one normalization, Gate 3 stops. Missing error propagation, physical-control failure, or an unavailable cohomology hypothesis map likewise stops the corresponding gate. These stops prevent an attractive determinant plot or internal identity from compensating for an upstream type error.

The architecture therefore treats provenance as part of the mathematical claim. A determinant value without the roof hash, operator definition, coefficient convention, error contract, and upstream receipts is not an incomplete instance of the registered result; it is a differently specified object. Conversely, an auditable stop does not prove that no suitable roof or operator could ever exist. It records exactly which registered evidence obligation remains unmet. This bounded interpretation is what makes the proposed workflow useful without converting it into a theorem or experiment.

2 Frozen Literature and Theoretical Frame

2.1 A determinant-type firewall

The classical three-hard-disk record anchors physical geometry, trapped trajectories, collision coding, and geometric lengths, but any formula-level use remains bound to correction companion DOI 10.1063/1.457669; it does not certify a project roof ledger or transfer determinant [10].

The companion semiclassical study organizes periodic-orbit and zeta constructions and is subject to the same correction DOI for affected use, yet its spectral object is not the classical physical-roof determinant [11].

The exact-quantum scattering determinant is a further typed comparator, with affected use bound to correction companion DOI 10.1063/1.457670; it cannot be substituted for a classical transfer determinant [12].

Three-disk periodic-orbit quantization supports cycle organization in chaotic scattering without proving nuclearity or a project error contract [6].

Periodic-orbit expansions for classical smooth flows provide adjacent organizational machinery but no convergence or coefficient certificate for the frozen roof and cutoffs [7].

A broad treatment of hyperbolic n -disk scattering clarifies symbolic and determinant-type distinctions while leaving quantum trace-class conclusions unavailable to an unconstructed classical roof operator [25].

Rigorous work on waves outside several convex obstacles supplies geometric and periodic-ray context but not the project's basis, coefficient window, or numerical tolerance [15].

A primary proceedings record links several-body trajectories, zeta functions, and scattering poles, yet its source type and support boundary do not yield physical-roof fidelity or a certified determinant [16].

These distinctions motivate the registered firewall: common geometry does not transfer credit across differently typed determinants.

2.2 The exact $d = 6a$ object map remains prospective

Symbolic dynamics for hyperbolic flows provides a general interface among a Markov coding, return map, and roof, but it does not identify the exact three-disk collision section or pointwise flight function [4].

General Axiom-A flow and suspension theory gives a broad thermodynamic frame without automatically establishing applicability to the open-billiard repeller under the project's conventions [5].

The closest frozen open-billiard transfer-operator source is conditional on explicit convexity, visibility, normalization, coding, roof, and function-space hypotheses that must be mapped to the exact object [23].

Spectral estimates for Ruelle transfer operators of Axiom-A flows are likewise conditional and do not choose the project's section, regularity class, basis, rank, or cutoffs [24].

Generalized zeta meromorphic continuation provides an adjacent analytic component but no pointwise $d=6a$ roof or lawful finite coefficient comparison [19].

Analytic-flow zeta machinery is similarly bounded by its isolated-set and analyticity hypotheses, which have not been discharged for the frozen project object [9].

The smooth-flow zeta record is correction-sensitive: the frozen verification permits its first meromorphicity part but requires the first-party erratum for affected Section-7 use [13].

The erratum is therefore a mandatory boundary for that affected use and does not add physical-roof evidence or independent peer-reviewed support [14].

An exact future object map would need to connect $d=6a$ geometry, collision section, symbolic states, inverse branches, pointwise Euclidean-flight roof, roof regularity, potential and normalization, operator space, determinant convention, and complex domain. Shared category labels such as hyperbolic, Axiom A, and open billiard cannot replace field-by-field hypothesis matching. Fitting the roof from inherited orbit totals is excluded because it would reverse the intended physical construction.

2.3 Roof-agnostic internal calibration

Periodic zeta and transfer identities provide the formal background for comparing consistently typed orbit/repetition weights, but physical specificity is not part of that algebraic consistency [20].

Fredholm-determinant machinery can support analytic identities for eligible operators under stated hypotheses, while it cannot establish nuclearity or the physical identity of an unconstructed project operator [21].

The present article treats Euler-product, periodic-trace, and Fredholm-determinant agreement as a design principle with three levels. At the formal level, coefficient expressions can be compared only after their roof, repetitions, signs, and normalization are typed consistently. At the analytic level, a determinant identity additionally requires a specified operator, space, domain, and convergence theorem. At the finite numerical level, agreement is an implementation calibration on a declared window and tolerance. None of these levels proves that the roof is the physical Euclidean-flight roof. Unit, shuffled, neighboring-geometry, and physical roofs may each be internally coherent when constructed consistently.

2.4 Four numerical components and a separate input-uncertainty channel

Explicit transfer-operator eigenvalue bounds can inform a future orbit- or spectrum-tail component only after the operator space and constants are instantiated [1].

Eigenvalue estimates on holomorphic function spaces offer another conditional component, not proof that the three-disk transfer operator acts on the required domain [2].

Spectral-Galerkin analysis for uniformly expanding maps supplies adjacent projection-error ideas but requires a lawful comparison to the flow operator and chosen basis [26].

Numerical Fredholm-determinant analysis supports evaluation-error reasoning for eligible trace-class integral operators while leaving trace class, coefficient correspondence, and project-specific error composition unproved [3].

The prospective error object is therefore a five-entry ledger rather than an asserted scalar identity:

```
E_geometry/roof-input
E_orbit-tail
E_rank/projection
E_quadrature/evaluation
E_roundoff
```

The last four are numerical approximation components. The first is separately typed input or model-enclosure uncertainty. A defensible enclosure would require a declared roof norm, an operator norm, and an output norm; a stability statement from roof perturbation to operator perturbation; a legal coefficient map between orbit and projected-operator representations; propagation of all five bounds through the chosen approximation chain; and a determinant-conditioning estimate on the declared complex domain. Without these links, even individually small terms cannot be called additive, exhaustive, or complete. The frozen corpus supplies no dedicated roundoff theorem, no geometry-input theorem, no common norm, and no determinant-conditioning bound for the project.

2.5 Directional Livšic reasoning

Periodic-data cohomology theory supports a negative obstruction under stated hyperbolic or symbolic hypotheses, but finite periodic agreement cannot prove a global coboundary relation [18].

Regularity and perturbative foundations for the Livšic equation refine the required hypothesis map without replacing all-periodic-data obligations by a finite panel [8].

Matrix-cocycle theory is stronger and differently typed than the scalar roof comparison, so it is background rather than a project-specific nontransfer result [17].

A current abelian extension applies under its own Anosov-flow hypotheses and does not automatically cover the open three-disk repeller or license a positive conclusion from a finite cutoff [22].

The admissible inference is asymmetric. After exact system and theorem hypotheses have been matched, one certified periodic-sum mismatch may obstruct one predeclared relation of the form $\rho_1 = c \rho_2 + h - h \circ \sigma$, with $c > 0$. Finite agreement does not establish the equality for all required periodic orbits. A mismatch against one scaling relation does not prove every form of inequivalence. No project-specific mismatch or equivalence result is reported here.

3 Executed Methodology: Frozen Evidence Synthesis Only

The executed research method was the already completed bounded literature workflow. Phase 2 inspected 68 record manifestations, removed 16 duplicate manifestations, screened 52 unique records at title or abstract level, and admitted 26 records after scope screening. Twenty-four of the 26 were counted in the workflow's peer-reviewed-journal numerator. The source inventory contains exactly P30-S01 through P30-S26. Independent verification recorded 26 VERIFIED existence outcomes and exact core metadata after bounded normalization. The present composition did not repeat or extend this retrieval.

Three Gaspard–Rice correction-companion omissions were previously repaired and independently rechecked without adding source IDs: P30-S01 and P30-S02 remain bound to DOI 10.1063/1.457669, and P30-S03 remains bound to DOI 10.1063/1.457670. P30-S17 remains bound to P30-S18 for the affected Section-7 boundary. The frozen reference inventory has no separate entries for the first two companion corrections because neither new source IDs nor altered closure were authorized; correction provenance therefore remains publication-incomplete even though the workflow bindings are visible.

Phase 3 encoded all 26 sources in a source-effect matrix and synthesized six thematic groups and six candidate compatibility tensions. Phase 4 compiled that evidence. Four procedurally separated Phase-5 roles then reviewed the report, followed by a synthesis and checkpoint. The present manuscript integrates only the author-adjudicated changes registered in the Phase-6 manifest. No new webpage, database, publisher record, full text, passage locator, quotation, scientific artifact, or result was retrieved or created.

The evidentiary ceiling is explicit. All 26 inherited prose citation pairs in this article retain `anchor: none`; all 144 inherited citation pairs across the five-paper Round-10 batch likewise lack passage locators. Source identity, DOI resolution, metadata checks, and bounded scope notes do not establish claim-to-passage support. Every source-bearing premise therefore has claim-to-passage status `INCONCLUSIVE`.

No direct quotation appears. Structured retraction status remains NOT_CHECKED and source-level conflict-of-interest status remains UNKNOWN_NOT_AUDITED for all 26 records.

3.1 From source roles to bounded method claims

The Stage-2 transformation follows the registered claim plan rather than searching for a stronger narrative. Each claim carries a frozen source set and a negative constraint. Physical-scattering sources can motivate geometry and coding only within their correction bindings. Symbolic-flow and operator sources supply conditional components only under their own hypotheses. Determinant and numerical-analysis sources can describe what an eligible approximation might require, but they cannot establish eligibility for an operator that has not been defined. Cohomology sources license only directional reasoning after an exact theorem-to-object map. The manuscript preserves these roles instead of merging them into a synthetic theorem.

Metadata verification is not passage verification. All inventory records have verified existence and normalized core metadata within the frozen workflow, yet the manuscript has no theorem, page, section, or paragraph locator for any cited premise. Citation comments therefore record a uniform inconclusive claim-to-passage state. Correction provenance is kept separately: the three Gaspard–Rice records retain their companion-DOI obligations, and P30-S17 remains paired with P30-S18 for affected use. This handling prevents an uncorrected formula transfer but does not amount to a structured retraction or full integrity audit.

The closed-corpus method yields a reviewable architecture and a complete 26-entry bibliography, not a scientific object. It did not generate collision coordinates, evaluate a flight segment, choose a section, define a Banach or Hilbert space, construct an operator, calculate a trace, approximate a determinant, or compare periodic sums. Rephrasing a conditional component in a coherent manuscript cannot advance any gate. This explicit execution boundary allows a later implementation to cite the present work as a design contract while preventing the contract from being mistaken for an implementation receipt.

4 Review-Adjudicated Method Architecture

4.1 Gate 1: pointwise physical-roof construction

A future certificate must freeze the collision section, symbolic states, inverse branches, geometry evaluator, coordinate and precision conventions, no-eclipse checks, and a pointwise Euclidean-flight roof derived independently of inherited periodic totals. It must state the roof’s regularity and serialize interval-valued input uncertainty. If the necessary collision geometry is unavailable or the pointwise map cannot be constructed without fitting, the gate returns PHYSICAL_ROOF_NOT_EVALUABLE. No roof is constructed here.

4.2 Gate 2: operator eligibility

The roof must be attached to one explicitly normalized transfer operator acting on one declared function space over one complex domain. Every theorem hypothesis must map to a project field, including expansion or hyperbolicity conditions, smoothness or analyticity, nuclear or trace property, and boundary convention. Missing fields produce OPERATOR_NOT_EVALUABLE; shared terminology does not confer eligibility. No operator or function space is selected here.

4.3 Gate 3: common coefficient map

Orbit cutoff and finite-rank projection are differently typed approximations. A prospective proof must identify which complete primitive-and-repetition coefficient through order N corresponds to which operator trace, projected trace, and determinant coefficient under the same roof and normalization. Agreement of array indices, truncation integers, or similarly named coefficients is not a legal map. If no comparison theorem is available, the correct state is `CALIBRATION_NOT_EVALUABLE`. No coefficient comparison is proved here.

4.4 Gate 4: five-channel common-norm error theorem

The proposed gate has four numerical components and a separate geometry/roof-input component. It is a proof obligation, not a completed theorem. A future certificate must specify:

1. a bound for the physical geometry and pointwise-roof input in a declared roof norm;
2. an orbit-tail bound for the complete orbit-and-repetition representation;
3. a rank or projection bound for the selected operator space and basis;
4. a quadrature or determinant-evaluation bound on the declared complex domain; and
5. a roundoff bound tied to arithmetic precision, algorithms, and stopping rules.

These bounds cannot simply be summed. The proof must supply a stability map from roof norm to operator norm, a propagation map through coefficient extraction and projection, and a conditioning estimate from operator perturbation to the determinant quantity. It must also account for dependencies or overlap among channels. Only the resulting composed inequality, with constants and domain explicit, could justify an enclosure. If any channel, norm, map, or conditioning factor is absent, the state is `ERROR_CONTRACT_NOT_EVALUABLE`. The present article reports no complete total error and no numerical bound.

4.5 Prospective composition of the five error channels

The five-entry ledger is a typing device before it is a numerical formula. Geometry and roof-input uncertainty begins with the declared disk parameters, collision coordinates, section conventions, and the interval or exact arithmetic used by the flight evaluator. Its output is a set or bound for pointwise roofs in a stated roof norm. Orbit-tail error concerns omitted primitive cycles and repetitions in a particular periodic representation. Rank or projection error concerns replacement of an infinite-dimensional operator by a finite-dimensional one. Quadrature or evaluation error concerns how the selected finite object is integrated or how its determinant is evaluated. Roundoff concerns the arithmetic operations, precision schedule, and stopping rules actually used. These channels enter at different stages and cannot share a unit until connecting maps are supplied.

A lawful composition would begin with a stability statement that transports a roof perturbation into an operator perturbation on the declared function space. It would then relate the complete periodic representation to the operator coefficients under one roof and normalization. Projection and evaluation maps would carry their own operator or coefficient norms. Finally, a determinant-conditioning statement would convert the accumulated perturbation into a bound for the reported determinant quantity over the declared complex domain. Every constant would be attached to a hypothesis record and a domain of validity. A scalar displayed beside five tolerance labels would not provide these maps.

Dependencies require special treatment. Geometry uncertainty can change every orbit weight coherently, so it may overlap with what a naive orbit-tail estimate treats as coefficient uncertainty. Projection and quadrature errors may share evaluations or basis data. Roundoff may be amplified by the same

ill-conditioning that controls determinant sensitivity. Consequently, addition by the triangle inequality would need a common ambient norm and verified coverage, while sharper combination would need an explicitly justified dependency model. The architecture prescribes neither numerical independence nor quadrature of errors. It asks a future proof to state which dependence assumptions, if any, it uses.

The error gate also distinguishes local calibration from uniform enclosure. Agreement at selected parameter values can test software paths, and a precision ladder can expose unstable digits, but neither observation proves a bound on a complex domain. A prospective certificate must state whether it encloses coefficients, traces, zeros, determinant values, or another output, because their condition numbers differ. It must also specify the domain boundary and behavior near zeros or spectral features. No such domain, output tolerance, or conditioning constant is reported here; the discussion fixes the questions that a later calculation must answer.

4.6 Gate 5: physical fidelity and typed controls

Internal determinant identities test consistency, not physical identity. A prospective physical roof must reproduce independently geometry-derived periodic flight times within a priori input and numerical errors. Unit-roof, pointwise-shuffled-roof, and neighboring-geometry controls must be constructed as actual roofs on the same coding interface. A shuffle of finite orbit labels is not a pointwise roof and cannot be promoted into an operator control. All choices and tolerances must be frozen before comparison. No fidelity or control result exists here.

4.7 Gate 6: theorem-licensed nontransfer

The final gate may use a certified mismatch only after matching the exact scalar cohomology theorem hypotheses and predeclaring the relation under test. It may support a negative obstruction to that relation; it cannot infer every inequivalence, and finite agreement cannot prove global equivalence. If the system or theorem map is incomplete, the state is `NONTRANSFER_NOT_PROVED`. No nontransfer witness is executed here.

4.8 Gate dependencies, artifacts, and independent replay

The six gates form a dependency graph rather than six interchangeable tests. Gate 1 emits the geometry and pointwise-roof ledger. Gate 2 consumes that exact ledger and emits an operator specification with a space, domain, potential, normalization, and verified hypothesis map. Gate 3 binds orbit coefficients, traces, projected traces, and determinant coefficients under the same specification. Gate 4 consumes those bindings to produce, if possible, one domain-specific enclosure. Gate 5 compares the physical roof with independently constructed controls while preserving the same coding interface. Gate 6 is optional and directional: it consumes a separately registered pair of roofs and an applicable scalar cohomology theorem map. A later gate cannot retroactively validate an earlier one.

Every artifact would need deterministic serialization and a content hash. A replay process would re-evaluate collision geometry, check that no periodic total was used to fit the pointwise roof, verify operator and coefficient metadata, reproduce error-bound arithmetic, and confirm that controls are actual pointwise roofs rather than shuffled summary labels. It would also confirm that any mismatch witness refers to the predeclared relation and that the theorem hypotheses are attached to the same system. Replay agreement would establish reproducibility of the registered finite procedure, not physical truth beyond its proved object and error scope.

The stop-state ledger is therefore substantive. It tells a reader whether the obstruction is geometric input, operator applicability, coefficient comparability, error composition, physical fidelity, or cohomological licensing. Reporting the first failed gate prevents downstream quantities from being interpreted

under a stronger type. In the present article, all six gates remain open because none has been executed. Their ordering and required artifacts are the methods contribution; they are not proxy results for a roof, determinant, enclosure, or nontransfer theorem.

5 Evidence-Synthesis Findings

The first registered finding is a type boundary. The physical flow, semiclassical periodic-orbit constructions, exact-quantum determinants, and classical transfer determinants remain distinct even when they share geometry or coding. Evidence for one object cannot be transferred to another without an exact map.

The second finding is that the $d=6a$ physical-roof certificate requires a complete object map from collision geometry through roof regularity and normalization to an eligible operator space and determinant convention. The corpus supplies conditional components, not that instantiated map. Fitting a pointwise roof from inherited periodic totals remains prohibited.

The third finding is that Euler, periodic-trace, and Fredholm-determinant agreement is roof-agnostic internal calibration for a consistently typed roof. It is not evidence that the roof is physical. Formal, analytic, and finite numerical levels must remain separate.

The fourth finding is the revised error architecture. Four numerical components must be supplemented by separately propagated geometry/roof-input uncertainty. A common norm, stability, propagation, and determinant-conditioning contract is mandatory before any combination. The article does not assert a complete total error or claim that the frozen sources close any of those project-specific obligations.

The fifth finding preserves the directional asymmetry of Livšic-type reasoning: a certified mismatch may obstruct one registered relation after hypothesis matching, while finite agreement cannot establish global equivalence. The sixth finding organizes these boundaries into six open gates. The seventh identifies the article as a falsifiable physical-determinant methods and error-contract contribution, not a computed determinant or arithmetic candidate result. The eighth preserves the Route boundary without change.

6 Reproducibility and Prospective Interface

Manuscript-level reproducibility is bounded by named immutable inputs: the Phase-2 source inventory and verification records, the Phase-3 source-effect matrix and synthesis, the Phase-4 report and claim manifest, all four Phase-5 role reviews, the synthesis checkpoint, and the Phase-6 claim-intent manifest. They preserve record identity, correction relations, source-role coding, prohibited transfers, and review findings. They do not provide exact passage locators or a complete external-search replay, and the present article does not claim that they do.

A future scientific package would require deterministic schemas for geometry, roof, operator, coefficients, errors, and controls. They would record a , $d=6a$, the section and collision inputs; pointwise evaluator bytes and roof uncertainty; operator space, domain, potential, basis, and determinant convention; orbit-to-trace coefficient bindings; all five error channels with norm, stability, propagation, dependency, and conditioning witnesses; and typed physical, unit, shuffled, and neighboring-geometry roofs.

Independent replay would check hashes, types, completeness, precision ladders, and stop states under prospectively frozen rules. All of these interfaces remain prospective. No serializer, proof object, implementation, or replay receipt is claimed to exist at Stage 2.

7 Discussion and Implications

Physical fidelity begins before numerical approximation. Accurate evaluation for an approximate roof can still miss the intended physical determinant when the operator is sensitive to roof perturbations. Geometry uncertainty must therefore travel through explicit stability and conditioning maps rather than being hidden inside quadrature or roundoff.

Internal agreement is necessary but insufficient: unit and physical roofs may each support coherent formulas. Agreement can expose within-object coding or normalization errors, but geometry-derived witnesses identify the Euclidean-flight object. The six typed gates make a later classical-dynamics computation auditable, while their nonarithmetic methods value creates no Route-A arithmetic relevance.

8 Acknowledged Limitations

The source corpus is bounded and not a novelty review. It cannot support a claim that no exact object map, roundoff theorem, or comparable workflow exists outside the frozen search. Twenty-four records predate 2021 because foundational geometry, dynamics, operator, numerical, and cohomology sources dominate the admitted corpus. Source age and search boundaries cannot establish originality.

All 26 paper-specific citation pairs and all 144 batch citation pairs lack exact passage locators. Claims have been narrowed to the frozen source-role summaries, but claim-to-passage support remains INCONCLUSIVE. No formula, theorem, section, paragraph, or quotation has been cleared for project-specific transfer. The Gaspard-Rice correction companions remain mandatory but lack separate frozen source IDs and full reference entries. P30-S17/P30-S18 must remain paired for affected use. Retraction status is not checked, and source-level conflicts are not audited.

No pointwise roof, exact section, operator, function space, coefficient map, rank, quadrature, precision ladder, determinant, error bound, physical period validation, control roof, or nontransfer witness is constructed or evaluated. The corpus does not supply a dedicated project roundoff theorem, geometry-input theorem, common norm, roof-to-operator stability estimate, propagation theorem, or determinant-conditioning bound. The five-channel contract is therefore a prospective architecture, not a complete total error.

The source retains nonprinting citation-provenance comments and a compact status ledger because the Stage-2 audit contract requires them. No independently reproducible search supplement was authorized. The article type is fixed as methods and evidence architecture, while contribution originality remains unassessed.

Any later execution package should expose the smallest evidence unit that permits independent failure diagnosis. The geometry layer should carry disk and section conventions, collision points, interval inputs, and pointwise flight evaluations. The operator layer should carry the potential, normalization, space, complex domain, and field-by-field hypothesis map. The coefficient layer should record primitive words, repetitions, trace conventions, projection indices, and the exact comparison theorem being used. The error layer should preserve every raw bound before composition, including its norm, constants, dependencies, and valid domain. The control layer should serialize physical, unit, shuffled, and neighboring-geometry roofs through the same callable interface.

That package should also distinguish validation from selection. A control may reveal that an implementation reacts to roof changes, but it may not be chosen after the physical determinant is inspected. A precision ladder may diagnose roundoff, but it does not replace a roundoff proof or a determinant-conditioning bound. A finite periodic panel may expose a mismatch, but its composition and the relation under test must be frozen before the mismatch is used. Finally, replay should fail if any upstream hash changes, even when the numerical output changes only slightly. These requirements make later progress

reportable gate by gate without allowing a downstream numerical success to rewrite the physical object.

9 Future Work

The smallest source step is a separately authorized locator pass for open-billiard operator, determinant, approximation, correction, and scalar-cohomology premises, followed by a roundoff and roof-input propagation strategy. The smallest object step is a field-by-field $d=6a$ map from geometry to a pointwise roof and eligible operator.

Only after that map is frozen should a common coefficient theorem and five-channel common-norm proof be attempted. A scholar-confirmed design would then register typed roofs, cutoffs, precision, domains, stop states, and replay rules. Any nontransfer test must remain directional and theorem-licensed.

10 Conclusion

The frozen corpus supports a rigorous program for asking whether a classical determinant belongs to the physical $d=6a$ three-disk flow, but it does not supply the determinant. The present article preserves the type firewall, turns roof-agnostic calibration into an explicit design principle, and replaces an under-specified four-part total-error claim with a five-channel proof obligation: four numerical components plus separately propagated geometry/roof-input uncertainty under one norm, stability, propagation, and conditioning contract.

All six scientific gates remain open. The defensible output is a falsifiable methods and evidence architecture, not a roof, theorem, computation, enclosure, fidelity result, nontransfer certificate, or novelty finding. The Route state remains `AO_FAIL` / `A2_NOT_ELIGIBLE` / `NO_ROUTE_PROMOTION`, the formal tuple remains `UNASSIGNED`, positive arithmetic `A2` remains `0/1`, and Route B remains uninvoked.

The architecture makes a future negative outcome interpretable. A roof-construction stop differs from an operator-applicability stop; an error-contract failure differs from a physical-fidelity failure; and a missing theorem map differs from a certified periodic-sum obstruction. Preserving those distinctions is the present contribution. Until separately authorized implementations and replayable certificates exist, none of the six gates may be treated as partially passed by prose alone.

Accordingly, the manuscript supplies a complete prospective decision structure while leaving every physical, analytic, numerical, and cohomological output explicitly unevaluated.

Author Contributions

Liang Wang is the responsible human author. CRediT-style roles: Conceptualization; Methodology; Writing—review and editing; Project administration. He supplied the conceptual direction, made the author-adjudication decisions, approved the recorded workflow gates, interpreted the bounded evidence synthesis, and accepts final accountability for the manuscript. AI assistance is disclosed separately and is not credited with authorship.

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Competing Interests

The author declares no competing interests.

Ethics Statement

Ethics approval, informed consent, and human-subject review are not applicable. This article concerns mathematical objects, published-source records, and project-owned workflow artifacts; it reports no human participants, intervention, identifiable personal data, or human-related dataset.

Data and Materials Availability

The evidentiary materials are the frozen project artifacts identified in the manuscript. No new collision data, pointwise roof, operator artifact, determinant value, numerical bound, fidelity comparison, cohomological witness, or experimental dataset was generated for this Stage-2 composition.

AI-Assistance Disclosure and Verification Limitation

OpenAI Codex, operating in the GPT-5 model family, assisted during the session dated 2 September 2026 UTC; the exact backend snapshot or build was not exposed. Assistance covered organization of the frozen bibliography and source identities, bounded evidence synthesis, drafting, review integration, LaTeX conversion, and deterministic checks of identifiers, citations, references, hashes, and word counts. No AI-authored scientific data, physical roof, operator, proof, determinant, error bound, cohomology result, or numerical experiment is presented. AI assistance is not credited with authorship.

Liang Wang made the recorded stage-gate confirmations and author-adjudicated design decisions. Neither that responsibility nor the present composition implies human full-text or source-passage verification. Verification remains limited to frozen metadata, abstracts or authorized scope notes, source-verification ledgers, the source-effect matrix, and review artifacts. Exact passages were not frozen; all citation anchors therefore remain none, and every claim-to-passage assessment remains INCONCLUSIVE pending Stage 2.5.

References

- [1] Oscar F. Bandtlow and Oliver Jenkinson. Explicit a priori bounds on transfer operator eigenvalues. *Communications in Mathematical Physics*, 276(3):901–905, 2007. doi: 10.1007/s00220-007-0355-7. URL <https://doi.org/10.1007/s00220-007-0355-7>.
- [2] Oscar F. Bandtlow and Oliver Jenkinson. Explicit eigenvalue estimates for transfer operators acting on spaces of holomorphic functions. *Advances in Mathematics*, 218(3):902–925, 2008. doi: 10.1016/j.aim.2008.02.005. URL <https://doi.org/10.1016/j.aim.2008.02.005>.
- [3] Folkmar Bornemann. On the numerical evaluation of Fredholm determinants. *Mathematics of Computation*, 79(270):871–915, 2010. doi: 10.1090/S0025-5718-09-02280-7. URL <https://doi.org/10.1090/S0025-5718-09-02280-7>.
- [4] Rufus Bowen. Symbolic dynamics for hyperbolic flows. *American Journal of Mathematics*, 95(2): 429–460, 1973. doi: 10.2307/2373793. URL <https://doi.org/10.2307/2373793>.

- [5] Rufus Bowen and David Ruelle. The ergodic theory of Axiom A flows. *Inventiones Mathematicae*, 29(3):181–202, 1975. doi: 10.1007/BF01389848. URL <https://doi.org/10.1007/BF01389848>.
- [6] Predrag Cvitanović and Bruno Eckhardt. Periodic-orbit quantization of chaotic systems. *Physical Review Letters*, 63(8):823–826, 1989. doi: 10.1103/PhysRevLett.63.823. URL <https://doi.org/10.1103/PhysRevLett.63.823>.
- [7] Predrag Cvitanović and Bruno Eckhardt. Periodic orbit expansions for classical smooth flows. *Journal of Physics A: Mathematical and General*, 24(5):L237–L241, 1991. doi: 10.1088/0305-4470/24/5/005. URL <https://doi.org/10.1088/0305-4470/24/5/005>.
- [8] Rafael de la Llave, José Manuel Marco, and Roberto Moriyón. Canonical perturbation theory of Anosov systems and regularity results for the Livšic cohomology equation. *Annals of Mathematics*, 123(3):537–611, 1986. doi: 10.2307/1971334. URL <https://doi.org/10.2307/1971334>.
- [9] David Fried. Meromorphic zeta functions for analytic flows. *Communications in Mathematical Physics*, 174(1):161–190, 1995. doi: 10.1007/BF02099469. URL <https://doi.org/10.1007/BF02099469>.
- [10] Pierre Gaspard and Stuart A. Rice. Scattering from a classically chaotic repeller. *The Journal of Chemical Physics*, 90(4):2225–2241, 1989. doi: 10.1063/1.456017. URL <https://doi.org/10.1063/1.456017>. Formula-level use is bound to correction companion DOI 10.1063/1.457669.
- [11] Pierre Gaspard and Stuart A. Rice. Semiclassical quantization of the scattering from a classically chaotic repeller. *The Journal of Chemical Physics*, 90(4):2242–2254, 1989. doi: 10.1063/1.456018. URL <https://doi.org/10.1063/1.456018>. Affected use is bound to correction companion DOI 10.1063/1.457669.
- [12] Pierre Gaspard and Stuart A. Rice. Exact quantization of the scattering from a classically chaotic repeller. *The Journal of Chemical Physics*, 90(4):2255–2262, 1989. doi: 10.1063/1.456019. URL <https://doi.org/10.1063/1.456019>. Affected use is bound to correction companion DOI 10.1063/1.457670.
- [13] Paolo Giulietti, Carlangelo Liverani, and Mark Pollicott. Anosov flows and dynamical zeta functions. *Annals of Mathematics*, 178(2):687–773, 2013. doi: 10.4007/annals.2013.178.2.6. URL <https://doi.org/10.4007/annals.2013.178.2.6>. Affected Section-7 use is bound to P30-S18.
- [14] Paolo Giulietti, Mark Pollicott, and Carlangelo Liverani. Anosov flows and dynamical zeta functions (Errata), 2022. URL <https://arxiv.org/abs/2203.04917>. First-party preprint erratum.
- [15] Mitsuru Ikawa. Decay of solutions of the wave equation in the exterior of several convex bodies. *Annales de l’Institut Fourier*, 38(2):113–146, 1988. doi: 10.5802/aif.1137. URL <https://doi.org/10.5802/aif.1137>.
- [16] Mitsuru Ikawa. On zeta function and scattering poles for several convex bodies. In *Journées Équations aux Dérivées Partielles*, pages Article II, 1–14, 1994. doi: 10.5802/jedp.457. URL <https://doi.org/10.5802/jedp.457>.
- [17] Boris Kalinin. Livšic theorem for matrix cocycles. *Annals of Mathematics*, 173(2):1025–1042, 2011. doi: 10.4007/annals.2011.173.2.11. URL <https://doi.org/10.4007/annals.2011.173.2.11>.

- [18] A. N. Livšic. Cohomology of dynamical systems. *Mathematics of the USSR-Izvestiya*, 6(6): 1278–1301, 1972. doi: 10.1070/IM1972v006n06ABEH001919. URL <https://doi.org/10.1070/IM1972v006n06ABEH001919>.
- [19] Mark Pollicott. Meromorphic extensions of generalised zeta functions. *Inventiones Mathematicae*, 85 (1):147–164, 1986. doi: 10.1007/BF01388795. URL <https://doi.org/10.1007/BF01388795>.
- [20] David Ruelle. Zeta-functions for expanding maps and Anosov flows. *Inventiones Mathematicae*, 34 (3):231–242, 1976. doi: 10.1007/BF01403069. URL <https://doi.org/10.1007/BF01403069>.
- [21] David Ruelle. An extension of the theory of Fredholm determinants. *Publications Mathématiques de l’IHÉS*, 72:175–193, 1990. doi: 10.1007/BF02699133. URL <https://doi.org/10.1007/BF02699133>.
- [22] Richard Sharp. Abelian Livšic theorems for Anosov flows. *Bulletin of the London Mathematical Society*, 58(1):e70258, 2026. doi: 10.1112/blms.70258. URL <https://doi.org/10.1112/blms.70258>.
- [23] Luchezar Stoyanov. Spectrum of the Ruelle operator and exponential decay of correlations for open billiard flows. *American Journal of Mathematics*, 123(4):715–759, 2001. doi: 10.1353/ajm.2001.0029. URL <https://doi.org/10.1353/ajm.2001.0029>.
- [24] Luchezar Stoyanov. Spectra of Ruelle transfer operators for Axiom A flows. *Nonlinearity*, 24(4): 1089–1120, 2011. doi: 10.1088/0951-7715/24/4/005. URL <https://doi.org/10.1088/0951-7715/24/4/005>.
- [25] Andreas Wirzba. Quantum mechanics and semiclassics of hyperbolic n -disk scattering systems. *Physics Reports*, 309(1–2):1–116, 1999. doi: 10.1016/S0370-1573(98)00036-2. URL [https://doi.org/10.1016/S0370-1573\(98\)00036-2](https://doi.org/10.1016/S0370-1573(98)00036-2).
- [26] Caroline Wormell. Spectral Galerkin methods for transfer operators in uniformly expanding dynamics. *Numerische Mathematik*, 142(2):421–463, 2019. doi: 10.1007/s00211-019-01031-z. URL <https://doi.org/10.1007/s00211-019-01031-z>.